

Preparing Activity: NAVFAC

Superseding
UFGS-14 21 13 (May 2016)

UNIFIED FACILITIES GUIDE SPECIFICATIONS

References are in agreement with UMRL dated April 2026

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ELECTRIC TRACTION FREIGHT ELEVATORS

08/25

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UNIFIED FACILITIES GUIDE SPECIFICATIONS

References are in agreement with UMRL dated April 2026

SECTION 14 21 13

ELECTRIC TRACTION FREIGHT ELEVATORS 08/25

NOTE: This guide specification covers the requirements for electric traction freight elevators.

Adhere to UFC 1-300-02 Unified Facilities Guide Specifications (UFGS) Format Standard when editing this guide specification or preparing new project specification sections. Edit this guide specification for project specific requirements by adding, deleting, or revising text. For bracketed items, choose applicable item(s) or insert appropriate information.

Remove information and requirements not required in respective project, whether or not brackets are present.

Comments, suggestions and recommended changes for this guide specification are welcome and should be submitted as a Criteria Change Request (CCR).

NOTE: All DoD facility designs which include elevators must comply with UFC 3-490-06, Elevators available at:
<https://www.wbdg.org/dod/ufc/ufc-3-490-06>

NOTE: For elevators installed in NAVY facilities, any editing of non-bracketed requirements in this specification requires approval through the NAVFAC FEC VTE Program Lead Certifying Official.

PART 1 GENERAL

1.1 REFERENCES

NOTE: This paragraph is used to list the publications cited in the text of the guide specification. The publications are referred to in the text by basic designation only and listed in this paragraph by organization, designation, date, and title.

Use the Reference Wizard's Check Reference feature when you add a Reference Identifier (RID) outside of the Section's Reference Article to automatically place the reference in the Reference Article. Also use the Reference Wizard's Check Reference feature to update the issue dates.

References not used in the text will automatically be deleted from this section of the project specification when you choose to reconcile references in the publish print process.

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A17.1/CSA B44 (2025) Safety Code for Elevators and Escalators

ASME A17.2 (2023) Guide for Inspection of Elevators, Escalators, and Moving Walks

ASME BPVC SEC IX (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41 (1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2024) International Building Code

NATIONAL ELEVATOR INDUSTRY, INC. (NEII)

NEII-1 (2000; R thru 2017) Building Transportation Standards and Guidelines, including the Performance Standards Matrix for New Elevator Installation

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2026; TIA 26-1; ERTA 26-1; TIA 26-2; TIA 26-3; TIA 26-4; TIA 26-5; TIA 26-6; TIA 26-7; ERTA 26-2; ERTA 26-3) National Electrical Code

NFPA 70E (2024) Standard for Electrical Safety in the Workplace

NFPA 72 (2025; TIA 25-1; TIA 25-2; TIA 25-3; TIA 25-4; ERTA 25-1; ERTA 25-2) National Fire Alarm and Signaling Code

NFPA 101 (2024) Life Safety Code

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-490-06 (2018; with Change 1, 2021) Elevators

UFC 4-010-06 (2023) Cybersecurity of Facility-Related Control Systems

UFM 3-560-01 (2026) Operation and Maintenance: Electrical Safety

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

1.2 SUBMITTALS

NOTE: For Navy and Army projects include the following requirement:

A Pre-submittal elevator design meeting must be conducted prior to submitting SD-02 Shop Drawings and SD-03 Product Data to review these proposed submittals. Coordinate the meeting with the Contracting Officer. Suggested attendees include the Contractor, elevator sub-contractor, FEC VTE LCO (or VTE SME), any involved design engineers, and any trades that are installing systems that interface with the elevator system.

NOTE: Review Submittal Description (SD) definitions

in Section 01 33 00 SUBMITTAL PROCEDURES and edit the following list, and corresponding submittal items in the text, to reflect only the submittals required for the project. The Guide Specification technical editors have classified those items that require Government approval, due to their complexity or criticality, with a "G." Generally, other submittal items can be reviewed by the Contractor's Quality Control System. Only add a "G" to an item if the submittal is sufficiently important or complex in context of the project.

For Army projects, fill in the empty brackets following the "G" classification, with a code of up to three characters to indicate the approving authority. Codes for Army projects using the Resident Management System (RMS) are: "AE" for Architect-Engineer; "DO" for District Office (Engineering Division or other organization in the District Office); "AO" for Area Office; "RO" for Resident Office; and "PO" for Project Office. Codes following the "G" typically are not used for Navy and Air Force projects.

The "S" classification indicates submittals required as proof of compliance for sustainability Guiding Principles Validation or Third Party Certification and as described in Section 01 33 00 SUBMITTAL PROCEDURES.

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Elevator System; G, [____]
Elevator Components; G, [____]
Wiring Diagrams; G, [____]
Elevator Drive Machine; G, [____]

SD-03 Product Data

NOTE: For Army projects, delete the bracketed items. For Navy projects, keep the bracketed items.

Elevator and Accessories[; G, [____]]
Elevator Components[; G, [____]]

Data Sheets[; G, [_____]]

Elevator Microprocessor Controller; G, [_____]

SD-05 Design Data

Emergency Power System

Heat Loads

Reaction Loads

SD-07 Certificates

Warranty

Welders' Qualifications

Elevator Mechanic State License, Certification, or Equivalent

SD-10 Operation and Maintenance Data

Elevator Controller Data Package; G, [_____]

Maintenance Control Program (MCP); G, [_____]

Software and Documentation; G, [_____]

SD-11 Closeout Submittals

Elevator Certification Certificate, Assessment, or Validation and Compliance Report; G, [_____]

1.2.1 Shop Drawing Requirements

Provide drawings showing assembly and arrangement of elevators, accessories, and elevator components with clearances marked. Provide floor plan and elevations showing location of all elevator equipment and supporting components including but not limited to: machine, controller, transformers, power panels, motors, disconnects, governors, oil coolers, Heating and AC units, storage cabinets, and human interface systems in the elevator machine room (MR) or control room (CR) and machinery spaces. Floor plans and elevations must be fully dimensioned including size of equipment and code required clearances for servicing components. Provide details for materials and equipment, including but not limited to operating and signal fixtures, doors, door and car frames, car enclosure, controllers, motors, guide rails and brackets, layout of hoistway in plan and elevation, and to indicate that other layout information and clearance dimensions have been met.

1.2.2 Product Data Requirements

Provide manufacturers' product data for all **elevator components**, including but not limited to the following: elevator controller, hoist machine and drive motor, design counterbalance, hoist ropes and shackles, overspeed governor, emergency braking system, car and hall fixture buttons and switches, cab, machine room, control room, and machinery space communication devices, door operator, door protection system, and car and

counterweight roller guides and buffers. For [data sheets](#), provide document identification number or bulletin number, published or copyrighted prior to the date of contract bid opening. Provide controller manufacturer's published procedures for performance of each and all testing required by [ASME A17.1/CSA B44](#).

1.2.3 Design Data

1.2.3.1 Reaction Loads

Provide calculations by registered professional engineer for [reaction loads](#) imposed on building by elevator system. Calculations must comply with [ASCE 7-16](#) and [ASME A17.1/CSA B44](#)

1.2.3.2 Heat Loads

Provide calculations from elevator manufacturer, or by registered professional engineer, for total anticipated [heat loads](#) generated by all of the elevator equipment.

1.2.3.3 Emergency Power Systems

Where the facility has an [emergency power system](#), confirm the elevators are connected to the emergency power system. Confirm the complete emergency power system and sequence of operation for all elevators, including elevator sequential operation and operation of the elevator lobby manual selection switch. Provide wiring diagrams for building emergency power interface with elevator controls. For elevators not supplied by an emergency power system, provide manufacturers' product data for auxiliary power systems.

1.2.4 Welders' Requirements

Comply with [ASME BPVC SEC IX](#). Include certified copies of field [welders' qualifications](#). List welders' names with corresponding code marks to identify each welder's welding work

1.2.5 Maintenance Control Program (MCP)

For each elevator, prepare and provide a written Maintenance Control Program (MCP) that complies with [ASME A17.1/CSA B44](#) Section 8.6 in its entirety. Include written documentation that details the test procedures for each and every test that is required to be performed by [ASME A17.1/CSA B44](#). Assemble all MCP documentation, and supporting technical attachments, in a single MCP package and provide in both electronic and hard copy. Assemble entire hard copy MCP in 3-ring binders and install in each elevator machine room (MR), machinery space (MS), control room (CR), or control space (CS). For each elevator provided, the MCP must include only documentation and instruction that apply to the installed elevator.

For each elevator, provide an additional, separate binder that includes all maintenance, repair, replacement, call back, and other records required by [ASME A17.1/CSA B44](#). Keep the records binder in the elevator MR, MS, or CR, maintained by elevator maintenance and service personnel, and available at all times to authorized personnel. Provide detailed information regarding emergency service procedures and elevator installation company personnel contact information. Provide a listing of all tools to be provided to the Contracting Officer as components of the

elevator system.

1.3 QUALITY ASSURANCE

1.3.1 Qualification of Elevator Personnel

Provide a designed and engineered elevator system by an elevator contractor regularly engaged in the installation of elevator systems. Provide elevator components manufactured by companies regularly engaged in the manufacture of elevator components. Utilize elevator personnel who are licensed, certified, or currently enrolled in an accredited elevator training program for adjusting, testing, and servicing of the elevators. Trainees, or helpers, are permitted to perform installation under the direct supervision of an elevator technician. Each elevator technician is permitted to supervise up to three trainees, or helpers.

1.3.1.1 Elevator Contractor's Elevator Technicians

For elevator installations in the United States, including United States territories, perform all elevator related work under the direct, and continuous daily on-site supervision of an elevator technician with a minimum of 3 years of experience in the installation of elevator systems of the type and complexity specified in the contract documents. All elevator technicians must provide a current [elevator mechanic state license, certification, or equivalent](#), and have documented completion of training provided by one of the following organizations:

- a. National Association of Elevator Contractors (NAEC)
- b. National Elevator Industry Education Program (NEIEP)
- c. National Association of Elevator Safety Authorities (NAESA)

Each elevator technician is allowed to supervise up to six trainees at any one time who are enrolled and actively participating in any of the above listed organization's training programs.

1.3.2 Manufacturers' Technical Support

Provide elevator components from manufacturers that provide factory training and online and live telephone elevator technical support to any elevator installation, service, and maintenance contractor. Provide elevator components from manufacturers that guarantee accessibility to all replacement and repair parts and components to any elevator installation, service, and maintenance contractor. Use only elevator component manufacturers that provide current published Price Lists for all elevator parts and components.

1.3.3 Operation and Maintenance Data

Assemble all shop drawing and product data material into O&M Data Packages in accordance with Article SUBMITTALS. Provide two complete O&M Data Packages in hard copy and two complete electronic O&M data packages on separate CDs, in PDF format. Provide all O&M Data Packages to Contracting Officer. Include controller diagnostic documentation and software as required under Article CONTROL EQUIPMENT.

1.3.4 Wiring Diagrams

Provide complete [wiring diagrams](#) and sequence of operations, which show

electrical connections and functions of elevator systems. Provide one set (279 mm by 432 mm 11 inch by 17 inch minimum size) of wiring diagrams, with individual sheets laminated in plastic and assembled in binder, stored in the machine room or control room cabinet. Provide one additional hard copy set and one complete electronic set in PDF format. Provide all wiring diagram sets to the Contracting Officer. Coded diagrams are not acceptable unless fully identified.

1.3.5 Machine Room/Control Room Cabinet

For storage of O&M Data Packages and Wiring Diagrams, provide locking metal cabinet with a minimum size of 508 mm W by 305 mm D by 762 mm H 20 inch W by 12 inch D by 30 inch H. Size cabinet large enough to accommodate all O&M Data, MCP, and hardware required in paragraphs OPERATION AND MAINTENANCE DATA and WIRING DIAGRAMS. Secure cabinet to machine room or control room wall.

1.4 NEW INSTALLATION SERVICE

NOTE: Use Bi-weekly option for Hospitals and other high use facilities.

Provide elevator warranty service in accordance with the manufacturer's maintenance plan, warranty requirements and applicable safety codes, for a period of 12 months after the date of acceptance by Contracting Officer. Perform this work during regular working hours. Provide supplies and parts to keep elevator system in operation. Perform service only by factory trained personnel. Provide[Monthly][Bi-weekly] services to include repairs, adjustments, greasing, oiling, and cleaning. Provide service log in elevator machine room or control room cabinet and update[Monthly][Bi-weekly], throughout the one-year warranty period.

NOTE: One hour emergency service below is standard; only use two hour for remote locations.

Provide 24-hour emergency service, with[one hour][two hour] on-site response time, during this period without additional cost to the Government.

1.4.1 Periodic Elevator Certification Inspection and Testing

Provide elevator mechanic to support[NAVFAC] QEI Certified Elevator Inspector in the periodic six-month and the annual Category 1 elevator certification inspection and testing. Perform Category 1 inspection and testing no greater than 30 days prior to the end of the warranty period. Perform all elevator certification testing in the presence of QEI Certified Elevator Inspector.

In conjunction with the testing noted above, test systems for Emergency Power Operation, Earthquake Emergency Operation, Fire Protection and Sprinkler Operation, and Hospital Emergency Commandeering Service Operation, as applicable. Schedule so that testing does not interfere with building operations.

1.5 FIRE PROTECTION SYSTEM

NOTE: Confirm that sections listed throughout this article are part of project. Add or delete sections as needed for project.

Coordinate interface between building fire protection system and elevator controls.

Additional fire protection requirements are located in:[Section 28 31 60 INTERIOR FIRE ALARM SYSTEM, NON-ADDRESSABLE,][Section 28 31 66 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, NON-ADDRESSABLE,][Section 28 31 70 INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE,][Section 28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE,][Section 28 31 32 FIRE ALARM REPORTING SYSTEM, DIGITAL COMMUNICATOR TYPE,][Section 21 13 13 WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION,] [_____] and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

1.5.1 Fire Alarm Initiating Devices

Fire alarm initiating devices are specified in[Section 28 31 60 INTERIOR FIRE ALARM SYSTEM, NON-ADDRESSABLE][Section 28 31 66 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, NON-ADDRESSABLE][Section 28 31 70 INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE][Section 28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE] [_____] , including conduit and wiring from each detector to fire protection addressable modules in elevator machine room or control room.

1.5.2 Fire Sprinklers

Provide fire sprinklers in accordance with all applicable safety codes and with[Section 21 13 13 WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION] [_____]. Provide shutoff valve, check valve, and non-adjustable, zero time-delay flow switch, in each sprinkler line immediately outside of each machine room, control room and hoistway, as applicable. Provide inspectors' test valve for periodic testing of flow switch and shunt trip disconnect.

Pipe sprinkler piping serving these spaces in a series manner with no laterals. Locate inspectors' test connection at the end of pipe runs such that operation of the test connection will purge air from system piping.

1.5.3 Shunt Trip Disconnect

Provide flow switches specified in paragraph FIRE SPRINKLERS to comply with ASME A17.1/CSA B44 and NFPA 72 for shunt trip of the main line power supply. For each elevator, provide control wiring connecting the flow switch to a shunt trip equipped circuit breaker located in the elevator machine room or control room. Upon flow of water, flow switch will instantaneously cause opening of the shunt-trip circuit breaker and remove power from the elevator. Flow switch must also send a signal to fire alarm control panel to indicate water flow condition.

PART 2 PRODUCTS

2.1 ELEVATOR DESCRIPTION

Provide elevator system that complies with ASME A17.1/CSA B44 in its entirety, UFC 3-490-06, and additional requirements specified herein. Provide elevator system that meets or exceeds the NEII-1 Ride Quality Performance Standards Matrix (RQPSM). For elevator speeds of 500 fpm and higher, comply with the RQPSM "High Performance" criteria. For elevator speeds 350 fpm, up to but not including 500 fpm, comply with the RQPSM "Intermediate Performance" criteria.

Provide and install elevators in accordance with 36 CFR 1191 - ABAAS, ICC IBC, IEEE C62.41, NFPA 70 and NFPA 101 requirements.

2.1.1 Elevator Design Parameters

NOTE: Traffic Analysis and Minimum Cab Size

Perform a traffic analysis and conduct interviews with the facility user to determine number, size, and type of elevators necessary to serve the needs of the facility user. Utilize UFC 3-490-06, Elevators to determine Design Type and speed.

NOTE: For Machine Roomless (MRL) installations, give careful consideration to the specific installation requirements in UFC 3-490-06, Elevators.

- a. Type: [Geared][Gearless]
- b. Configuration: [Overhead Traction][Basement Traction][Machine Roomless]
- c. Rated load: [_____] lb.
- d. Rated Speed: [_____] [150] [200] [350] [500] fpm
- e. Car Inside Dimensions: [_____] cm ft. [_____] cm in. wide, [_____] cm ft. [_____] cm in. deep and [_____] cm ft. [_____] cm in. high
- f. Hoistway Door Type & Size: [Manual][Power operated] Vertical [Rising][Bi-Parting] [_____] cm ft. [_____] cm in. wide and [_____] cm ft. [_____] cm in. high
- g. Car Gate Type: [Manual][Power operated] Vertical Rising
- h. Loading Type: Class [A][B][C]

2.1.2 Cab Enclosure and Hoistway Entrance Assemblies

NOTE: If retaining first option in sentence below, ensure that finishes are indicated, most likely somewhere on the drawings. In either case, indicate

finish colors of elevator materials in finish
schedule on drawings.

**NOTE: Specify stainless steel doors, side panels
and wall trim in hospital elevator cabs.**

Provide finishes[as indicated.][as listed below:

a. Floor;[mill finish steel diamond plate][painted steel diamond
plate][aluminum diamond plate][tongue and groove hardwood] [____].

b. Walls;[prefinished steel][stainless steel] [____].

(1) Wall trim;[prefinished steel][stainless steel] [____].

(2) Accessories;[hand rails] [____].

**NOTE: Retain bracketed phrase for freight elevators
that do not have Front and Rear openings.**

(3) Accessories; Provide in-car horizontal buck-board wall protection
on full length of side walls[and back wall] of elevator cab.

c. Car doors, car door returns, and wall reveals;[prefinished steel
panels][stainless steel] [____].

d. Ceilings;[supported][prefinished steel panels][anodized aluminum]
[____].

e. Hoistway Entrance Assembly Material and Finishes;[prefinished steel][
stainless steel] [____].

]2.2 ELEVATOR OPERATION

ASME A17.1/CSA B44, Introduction, Section 3, Definitions.

**NOTE: Choose one of the following four types of
elevator operation.**

[2.2.1 Single, Two-Stop, Automatic Operation

**NOTE: Choose for single elevator serving two
landings.**

Provide Single Two-Stop Automatic Operation.

]2.2.2 Selective Collective Automatic Operation

NOTE: Choose for single elevator serving three or

more landings.

Provide Selective Collective Automatic Operation.

][2.2.3 Duplex Selective Collective Automatic Operation

NOTE: Choose for two adjacent elevators.

Provide Duplex Selective Collective Automatic Operation. If a car is taken out of service or fails to respond to a landing call within a predetermined adjustable time limit of approximately 40 to 180 seconds, transfer calls to the other car functioning as a single car Selective Collective elevator until the out-of-service car is returned to the system.

][2.2.4 Group Automatic Operation

NOTE: Choose for three or more elevators that serve the same elevator lobby.

Provide Group Automatic Operation. If a car is taken out of service, or fails to respond to a landing call within a predetermined adjustable time limit of approximately 40 to 180 seconds, transfer calls to another car until out-of-service car is returned to the system.

][2.3 SPECIAL OPERATION AND CONTROL

Provide the following special operations and control systems.

2.3.1 Keys for Elevator Key Switches

Provide a minimum of three industry standard barrel keys per unique cylinder used on all key switches for a single elevator. If there is more than one elevator, provide one additional key set for each unit. Provide keys with plastic, or better, tags marked "PROPERTY OF THE U.S. GOVERNMENT" on one side with function of key on the other side. Provide two elevator drop keys for each elevator with red plastic tag stating 'Emergency Elevator Key (DANGER) For Use By Authorized Persons Only'.

2.3.1.1 Elevator Key and Switch Scheme

Fan speed, cab, and work lighting must be non-key operated type switch, located behind a locked service panel which is located on the COP. Follow the elevator key protocol as listed:

- a. EX511 Key operates the audible alarm reset/Loss of communication and is located at the fire recall level.
- b. EX512 Key opens the locked service panel door on the Car Operating Panel (COP) and operates the In-Car Stop switch located inside the locked service panel.
- c. EX513 Key operates the independent service switch and must be located on the COP accessible to the public (not behind the locked service panel door).

- d. EX514 Key operates the Inspection enable switch and the Access Enable switches located inside the locked service panel.
- e. EX516 operates the Hospital Commandeering/Car switches, if applicable, and is located accessible to the public on the COP, and at every landing hall call station.
- f. FEOK1 Key operates the Fire Service phase 1 switch located at the fire recall level and must open the fire service panel and operate the phase 2 switch located inside the firefighters service panel (FFS).

2.3.2 Firefighters' Emergency Operation (FEO)

NOTE: Coordinate FEO Designated Landing with Fire Protection Designer.

Provide FEO equipment and signaling devices. The designated level for the FEO Phase I key operated switch is the [ground][_____] floor. In the FEO Phase I fixture, provide FEO Operating Instructions.

2.3.2.1 Firefighters' Emergency Operation (FEO) Key Box

Provide a locking, FEO Key Box of a minimum inside dimensions of 137 mm W by 229 mm H by 35 mm D 5.375 inch W by 9 inch H by 1.37inch D. Install at a height of 1524 mm 5 feet above floor level and directly above the FEO Phase I key switch. Provide box equipped with lock that uses the FEO K1 key and sized large enough to house a set of elevator keys for each unit in the bank, and a standard drop key.

2.3.3 Hoistway Access Operation

Provide hoistway access operation with switches at top and bottom terminal landings. Locate switch 183 cm 6 feet above floor level, within 305 mm 12 inches of elevator hoistway entrance frame or with the ferrule exposed when located in the elevator entrance frame.

2.3.4 In-Car Inspection Operation

Provide In-Car Inspection Operation.

2.3.5 Independent Service

Provide key-operated switch on the car operating panel to enable independent service and simultaneously disable in-car signals and landing-call responses. Provide indicator lights that automatically illuminate during independent service. For duplex or group operation, if one car is removed from group another car will respond to its hall calls.

2.3.6 Selective Door Operation

For elevator with one or more rear openings at same level as front opening, provide full-selective operation with car and door operating buttons clearly marked for front and rear openings, front and rear car button for each such floor, and front and rear "DOOR OPEN" and "DOOR CLOSE" buttons. Only door for which the button was operated opens or closes.

[2.3.7 Elevator Emergency Power Operation

NOTE: Identify the elevators to be connected to the building emergency power system in the electrical design. Identify and define the complete emergency power system for all elevators. When using the second bracketed option in either of the next two subparagraphs, edit as required for project-specific requirements.

For any elevator that is not included in the building emergency power operation, utilize paragraph ELEVATOR AUXILIARY POWER OPERATING SYSTEM.

Provide elevator emergency power operation for[all elevators][elevator 1,2,3...]. Coordinate power supply and control wiring to accomplish initiation and operation of elevators on emergency power.

] [2.3.8 Elevator Auxiliary Power Operating System

Provide elevator auxiliary power operating system for[all elevators][elevator 1,2,3...].

] 2.4 ELEVATOR DRIVE MACHINE, HOIST MOTOR, AND DRIVE MOTOR

NOTE: Specify regenerative drive systems as an option only for installations which have greater than 80 feet of rise. Ensure coordination with electrical engineer before specifying a regenerative drive.

Provide elevator drive machine, hoist motor, and motor drive system that is designed to be installed in an elevator machine room (MR) or an elevator machinery space (MS).[Provide a regenerative drive type system.] The elevator machine, motor, and drive configuration and installation design must be mechanically and electrically interchangeable with a minimum of two other elevator manufacturer's drive machines that are readily available in the elevator industry. Paint or finish ferrous surfaces with a minimum of one coat of manufacturer applied rust-inhibiting paint.

2.4.1 Elevator Drive System Hoist Motor

Design the elevator drive system so that the hoist motor amperage does not exceed the motor data tag full load amperage in any operating condition, exclusive of acceleration and deceleration. Provide elevator hoist motor that is designed with Class F insulation and rated for a minimum of 120 starts/hour.

2.4.2 Elevator Drive System Suspension Means

Provide an elevator drive machine designed for and provided with stranded steel wire rope for elevator suspension and counterbalance. The minimum acceptable diameter of suspension and counterweight ropes is 9.52 mm 3/8

inches. Do not use aramid fiber ropes, coated steel ropes, and non-circular coated steel belts for elevator suspension or counterbalance.

2.4.3 Elevator Drive Machine Braking System

Equip the elevator drive machine with machine manufacturer's designed and installed standard means for the manual release of the driving-machine brake.

2.4.4 Manufacturer's Factory Training and Technical Support

Provide an elevator drive machine from a manufacturer that provides comprehensive factory training and technical support for installation, adjustment, service, and maintenance of the drive system. Identify the training and support as available to any licensed elevator contractor. Provide verification of an established and documented training schedule upon request, with pricing, for factory training classes that have been provided for a minimum period of one year prior to contract award date.

Identify the elevator drive system as available for purchase and installation by any licensed elevator contractor. All drive system related components, parts, diagnostic tools, and software must be available for purchase, installation, and use by any licensed elevator contractor; "exchange-only" provisions for the purchase of spare parts are not acceptable.

2.4.5 Ascending Car Overspeed and Unintended Car Movement Protection

Provide elevator Ascending Car Overspeed and Unintended Car Movement Protection means that is designed to act directly upon, and apply a retarding force to, the elevator suspension ropes or car rails. In addition to the requirements of ASME A17.1/CSA B44, the means must be designed to detect and stop movement of the elevator suspension ropes that occurs as a result of loss of traction between the suspension ropes and the elevator machine drive sheave.

2.5 CONTROL EQUIPMENT

Enclose all elevator control equipment in factory-primed and baked-enamel coated sheet-metal cabinets with ventilation louvers and removable or hinged doors. Mount cabinets at eye level to the User Interface Tool, with the maximum height of the cabinet top no greater than 200 cm 78 inches above the machine room or control room finish floor. For floor mounted controllers, maintain a minimum clearance of 254 mm 10 inches.

2.5.1 Motor Control Equipment

Provide variable voltage with silicon controlled rectifier (SCR) or Variable-Frequency Variable Voltage (VVF) alternating current (ac) drive control. Design the elevator drive system to limit Total Harmonic Distortion to a maximum of 5 percent. No single harmonic may exceed 3 percent.

2.5.1.1 Electrical Isolation Protection

Provide individual isolation transformers and individual choke reactors for each individual hoist motor. Provide filtering to maintain harmonic distortion below IEEE C62.41 standards as measured at the elevator machine room or control room disconnect.

2.5.2 Elevator Controller

For each individual elevator controller, and for each group controller, provide an **elevator microprocessor controller**. Provide **elevator controller data package** that includes all hardware and software required for the installation, maintenance, testing, and service of the elevator, in its entirety. Provide an elevator controller designed to automatically reestablish normal elevator operation following any temporary loss of power, regardless of duration. Remote Interaction Operation (RIO) is not permitted to be installed.

2.5.2.1 Elevator Controller Availability

Identify the elevator controller as available for purchase and installation by any licensed elevator contractor. All components, parts, diagnostic tools, and software must be available for purchase and installation and use by any licensed elevator contractor; "exchange-only" provisions for the purchase of spare parts are not acceptable. The elevator controller manufacturer must publish an industry competitive price listing for all controller parts, diagnostic tools, and software.

Provide an elevator controller from a manufacturer that provides comprehensive factory training to include controller installation, adjustment, service, and maintenance. Identify the training as available to any licensed elevator contractor. Training schedule, with pricing, for factory training classes that manufacturer provides must be made available upon request, and offered for a minimum period of one year prior to contract award date.

2.5.2.2 Elevator Controller Technical Support

Provide verification of telephone and internet based technical support service upon request, that the elevator controller manufacturer provides to any licensed elevator installation, service, and maintenance company at an industry competitive price. Include in the service live, telephone-based technical support for installation, adjustment, maintenance, and troubleshooting of the elevator controller and related elevator components. The service must be available during standard working hours.

2.5.2.3 Elevator Controller Interface Cabinet

For each individual elevator microprocessor controller, provide a separate elevator control cabinet with an integrated human interface system. For group elevator installations, a single cabinet and interface system with full access to each elevator controller may be utilized. The separate controller interface cabinet must be supplied by the elevator controller manufacturer and include a minimum **305 mm 12 inch** wide keyboard and a minimum **254 mm 10 inch** monitor. The elevator controller interface cabinet must comply with arc-flash protection requirements of **NFPA 70E** and **UFM 3-560-01**.

2.5.2.3.1 Elevator Microprocessor Human Interface

The interface system must provide complete elevator controller interface capability and must include the elevator controller manufacturer's comprehensive package of installation and diagnostic software. The interface cannot utilize wireless communication with the controller. The

microprocessor interface system must provide unrestricted access to all parameters, all levels of adjustment, and all flags necessary for installation, adjustment, maintenance, and troubleshooting of each elevator and for the elevator group. Store all software programming in non-volatile memory. The elevator controller fault log must provide non-volatile memory fault log storage of all faults, trouble calls, and fault history for a minimum of one hundred events, and the ability to download or print the fault log. The controller interface must also provide the capability to display and diagnose trouble calls, faults, and shutdowns. Expiring software, degrading operation, and "key" access controls are not acceptable.

2.5.2.4 Software and Documentation

Provide three copies of the manufacturer's maintenance and service diagnostic software, with complete software documentation, that enables the same level of unrestricted access to all controllers of the same make and model, regardless of the installation date or location. Manufacturer's maintenance and service diagnostic software that is available without charge for downloading via vendor portal meets this requirement. Revisions to software must be available at no additional charge for the life of the controller. Provide signed certification, from the manufacturer's corporate headquarters, that guarantees that the microprocessor software and access system will not terminate the unlimited and unrestricted access at any future date.

2.5.2.5 Elevator Controller Cybersecurity

Comply with [UFC 4-010-06](#), Cybersecurity for Facility Related Control Systems (FRCS), and Section [25 05 11](#) CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS, to ensure the control system is isolated and not accessible remotely. Controllers that have remote capability are acceptable if proven that their remote capability is permanently disabled.

2.6 OPERATING PANELS, SIGNAL FIXTURES, AND COMMUNICATIONS CABINETS

For all panels and fixtures, provide identical and uniform fixture design, material, finish, and components for all elevators. For all panels and fixtures, legibly and indelibly identify all buttons and all operating positions for each device. Use engraving and backfilling, or photo etching, for button and switch designations. Do not use attached signs. Provide elevator manufacturers' standard grade for all key switches unless otherwise specified. Utilize LED lighting, or equivalent, for energy efficiency for all illuminating panels and fixture components.

2.6.1 Car and Hall Buttons

For all cab and landing fixture buttons, provide industry-standard, vandal resistant push buttons with positive-stop assembly design. Buttons must be minimum [19 mm 3/4 inch](#) diameter, satin-finish stainless steel, with illuminating LED halo.

2.6.1.1 Hall Station Door-Operating Buttons

**NOTE: Use the following for freight elevators with
power-operated bi-parting doors.**

Provide hall station door-operating buttons, identical in size and design to hall call buttons, but not illuminating.

2.6.2 Freight Car-Operating Panel

NOTE: Use two Car Operating Panels for front and rear opening elevators.

Provide each car with[one][two] car operating panel[s] that contain[s] operation controls and communication devices. Provide exposed, flush mounted buttons for the controls identified in subparagraph PASSENGER CONTROLS. Provide a lockable service cabinet for the controls listed in subparagraph SERVICE CONTROLS. Use engraving and backfilling or photo etching for button and switch designations. Do not use attached signs.

2.6.2.1 Operator Controls

all passenger controls must be exposed and accessible to riders. In addition to ASME A17.1/CSA B44 requirements, provide the following operating controls, identified as indicated:

- a. LED illuminating car-call buttons identified to correspond to landings served by the elevator.
- b. "DOOR OPEN" and "DOOR CLOSE" buttons. For front and rear openings at the same floor, include the identification "F" and "R" for each opening.
- c. Red, illuminating "ALARM" button.
- d. Key-operated "Independent Service" switch.
- e. "Help" communication device to include intercom function, which when activated communicates between elevator cab and elevator machine room or control room.

2.6.2.2 Service Controls

Locate all service controls behind a locked cover and not accessible to riders without a key. In addition to ASME A17.1/CSA B44 requirements, provide the following operating controls, identified as indicated:

- a. Provide a key-operated, three-position switch for "In-Car Inspection Operation" and "Hoistway Access". The center switch position provides normal, automatic operation.
- b. "Car Light" must be a non-key operated switch.
- c. "Car Fan" must be a non-key operated switch with two speed settings identified.
- d. 120-volt ac 60 Hz single-phase duplex electrical outlet of ground-fault-circuit-interrupt (GFCI) design.
- e. Provide an In-Car stop switch located in the service cabinet.

2.6.2.3 Certificate Window

Provide a minimum 102 mm wide by 152 mm high 4 inch wide by 6 inch high certificate window for elevator inspection certificate. Locate window in the Service Controls door of the Car Operating Panel.

2.6.2.4 Emergency Signaling Devices

Provide an audible signaling device, operable from the Car Operating Panel button marked "ALARM". The audible signaling device must have a sound pressure rating between 80 and 90 dBA at 3 meters 10 ft. Provide battery backup power capable of operating the audible signaling device for at least one hour.

2.6.3 Elevator In-Car Position Indicators

For all elevators, provide illuminating LED position indicator in the Car Operating Panel.

2.6.4 Elevator In-Car Direction Indicators

For 2-stop elevator installations, provide visual direction indicators and audible car arrival signal in the elevator car door jamb, in accordance with ABA Standards. Visual indicators must be visible from the hall call fixture.

2.6.5 Hall Call Landing Fixtures

Provide a hall call fixture adjacent to each elevator. Provide a single push-button for terminal landings and dual push-buttons, up and down, at intermediate landings.

2.6.5.1 Designated Landing Hall Call Fixture

2.6.5.1.1 Location of Communication Means Failure (CMF) Visual Signal

As required by ASME A17.1/CSA B44, provide an elevator CMF audible and illuminating signal, and reset switch, in the FEO Designated Landing hall call fixture. Mount the signal and reset switch at a minimum of 178 mm 7 inches above the "UP" hall call button.

2.6.5.1.2 Communication Means Failure (CMF) Visual and Audible Signal Operation

Provide a CMF visual and audible signal system that conforms to ASME A17.1/CSA B44. Provide continuous verification of operability of the telephone line and within 60 seconds the activation of audible and visual signals when verification means determines that the telephone line is not functioning. Provide illumination of visual signal at one second intervals. Provide a minimum of 65 dBA audible signal at 30 second intervals.

2.6.5.1.3 Firefighters' Emergency Operation Phase I Switch and Visual Signal

When required by ASME A17.1/CSA B44, provide an elevator Firefighters' Emergency Operation Phase I switch and illuminating visual signal in the FEO Designated Landing hall call fixture. Provide FEO Phase I visual signal that is designed with intermittent, flashing, illumination when

actuated by the machine room, control room, or hoistway fire alarm initiating device. Locate FEO Phase I key switch above the CMF visual signal with a minimum of 152 mm 6 inches vertical between the centerlines of the CMF signal and the FEO Phase I key switch. Locate FEO Phase I visual signal directly above the Phase I switch. In addition, locate Elevator Corridor Call Station Pictograph at top of hall call fixture.

2.6.6 Elevator Car Position and Direction Indicators and Car Arrival Signal

For elevator installations with three or more stops, provide a separate hall landing fixture that includes the visual elevator position indicator, visual direction indicators, and audible car arrival signal, in accordance with ABA Standards.

2.6.7 Designated Landing Elevator Identification Fixture

For duplex and group elevator installations, install a separate elevator identification fixture for each elevator on the top center of the hoistway door jamb, with identification engraved and backfilled with a contrasting color. Number elevators from left to right, as seen during primary approach from building main entrance to elevator lobby. Lettering must be minimum of 19 cm 3/4 inch height. For multiple elevator groups, begin numbering with group that is closest to the building main entrance.

2.6.8 Emergency or Standby Power

When emergency or standby power is provided for elevator operation, provide an elevator emergency power visual indicator that conforms to ASME A17.1/CSA B44. Locate the visual signal in the Firefighters Emergency Operation fixture for each simplex elevator and for each elevator group. When an emergency power selector switch is required, provide switch in a separate, flush mounted fixture located at the designated level, in view of all elevator entrances.

2.7 CAR DOOR EQUIPMENT

2.7.1 Car Door Operator

Wireless communication systems are not permitted to be utilized.

2.8 FREIGHT ELEVATOR GUIDES, PLATFORM, AND ENCLOSURE

2.8.1 Guides

Provide coil-spring loaded roller guide assemblies in adjustable mountings on each side of car and counterweight frames in accurate alignment at top and bottom of frames. For freight elevators with a rated load greater than 10,000 lbs., slide guides may be used in lieu of roller guides.

2.8.2 Car Shell Return Panels, Doors, Entrance Columns, and Transom

Provide 14 Gauge minimum[prefinished steel][stainless steel] cab wall panels and entrance components. Use same material and finish for all hoistway and car entrance assemblies. Apply sound-deadening material on exterior of all cab wall panels.

2.8.3 Car Enclosure Top

Provide reinforced, 12 gauge minimum steel car enclosure top. Provide

hinged emergency exit with lock that complies with the appropriate seismic risk design requirements of [ASME A17.1/CSA B44](#). Locate emergency exit hinge towards the rear of the elevator cab. Design and configure the elevator cab interior ceiling to provide convenient and unobstructed access to, and use of, emergency exit from inside the elevator cab.

2.8.4 Car Door

Provide[two section] vertical rising gate with power door operator.

2.8.5 Car Entrance Sill

Provide one piece steel car entrance sill(s). Set sill(s) level and flush with cab finish floor.

2.8.6 In-Car Horizontal Buck-Board Wall Protection

**NOTE: Retain bracketed phrase for freight elevators
that do not have Front and Rear openings.**

Provide minimum 64 mm thick by 305 mm high 2-1/2 inch thick by 12 inch high #2 Oak protection boards on all side walls[and back wall] of the elevator cab. Position boards at a height to prevent damage from fork lift traffic.

2.8.7 Cab Finish Floor

Provide cab finish floor with top of finish floor flush with the cab sill.

2.8.8 Car Fan

Provide 2-speed fan for car enclosure forced ventilation. Mount fan in the car enclosure top.

2.8.9 Car Lighting

Utilize LED lighting for elevator car interior illumination. Provide a minimum of 10 foot-candles, measured at all areas of the car enclosure floor. Provide automatic car lighting operation that turns off car lights after 3 minutes of inactivity. Car lights must automatically turn on upon actuation of an elevator car or hall call.

2.8.10 Car Apron (Toe Guard)

Car aprons may be retractable, or adjustable, for existing pits that have limited clearance. Aprons must comply with deflection and deformation requirements of [ASME A17.1/CSA B44](#).

2.9 FREIGHT ELEVATOR HOISTWAY DOORS AND ENTRANCES

Provide fire rated hoistway entrance assemblies. Use same material and finish for all hoistway and car entrance assemblies.

2.9.1 Hoistway Entrance Frames

Provide 14 gage minimum[prefinished carbon sheet steel][stainless steel] hoistway entrance frames. Provide door panels with truckable sill.

Provide minimum 102mm width by 229 mm height 4 inch width by 9 inch height vision panel in upper door panel.

2.9.2 Hoistway Entrance Sills

Provide one-piece 8 mm thick, 152 mm by 152 mm 5/16 inch thick, 6 inch by 6 inch steel angle iron sill with top of sill flush with hoistway landing finish floor. Steel angle-iron sill must be set into and fully supported by the concrete floor or the steel building structure. Use same material for all hoistway and car entrance sills.

2.10 HOISTWAY EQUIPMENT

2.10.1 Car and Counterweight Guide Rails and Fastenings

Provide T-section type guide rails for car and counterweight. Paint rail shanks with one coat of black enamel.

2.10.2 Pit Equipment and Support Channels

Provide rail-to-rail pit channels to serve as mounting surface for main guide rails and counterweight guide rails. In addition, pit channels serve as mounting surfaces for car and counterweight buffers. Method of installation of channels, brackets and buffer mounts cannot puncture pit waterproofing. Paint all pit channels. Each car, or bank of cars, must have a set of portable barriers installed in the MR or CR for use in accessing any landing safely. Label barriers to indicate which car, or bank of cars, it is intended to be used for.

2.10.3 Pit "STOP" Switch

Provide push-to-stop/pull-to-run type pit "STOP" switch.

2.10.4 Traveling Cables

Suspend traveling cables by means of self-tightening webbed devices or internal suspension members.

2.10.5 Hoistway Pit Ladder

Provide continuous horizontal slip resistant rungs for the full height of the pit ladder. Center opening door systems require a retractable ladder to meet the required maximum distance from the hoistway door opening means. Ladder guarding is not permitted to be installed.

2.10.6 Pit Flooring Preservation

Paint pit floor with minimum 2 coats of gray industrial grade, high traffic rated paint.

PART 3 EXECUTION

3.1 INSTALLATION

Install in accordance with DOD design criteria, contract specifications, manufacturer's instructions, NEII-1 Building Transportation Standards and Guidelines, and all applicable building and safety code requirements.

3.1.1 Structural Members and Finish Materials

Do not cut or alter structural members. Do not alter finish materials from manufacturer's original design. Restore any damaged or defaced work to original condition.

3.1.2 Miscellaneous Requirements

Provide recesses, cutouts, slots, holes, patching, grouting, and refinishing to accommodate elevator installation. Use core drilling to drill all new holes in concrete. Finish work to be straight, level, and plumb. During installation, protect machinery and equipment from dirt, water, or mechanical damage. Paint all bare metal. At completion, clean all work and spot paint to match surrounding.

3.2 FIELD QUALITY CONTROL

NOTE: Use the bracketed item for all Navy Facilities and for all projects managed by NAVFAC FEAD, except as otherwise directed by the NAVFAC FEC VTE LCO. Use the bracketed item for Air Force and Army projects that are not managed by NAVFAC FEAD, unless utilizing the installing elevator contractor to perform pre-final inspection and testing.

The installing elevator Contractor must conduct elevator pre-acceptance inspection and testing[utilizing a currently licensed Qualified Elevator Inspector (QEI) not employed by the installing elevator contractor]. The inspections and testing ensures that the installation conforms to all applicable safety codes and contract requirements, meets all ASME testing requirements, and has passed all Field tests required by this specification. For modernization projects, existing equipment must be adjusted to meet performance criteria.

Upon completion submit a written test data report certifying that the elevator is complete and ready for final Acceptance Inspection, Testing, and Commissioning. Include all findings, measurements, and test results in the report.

3.3 ACCEPTANCE INSPECTION, TESTING AND COMMISSIONING

When elevator system installation is complete and ready for final inspection, notify Contracting Officer that elevator system is ready for Acceptance Inspection, Testing, and Commissioning.

NOTE: Use the first bracketed paragraph for all Navy Facilities and for all projects managed by NAVFAC FEAD. Use the second bracketed paragraph for Air Force and Army projects that are not managed by NAVFAC FEAD.

[Contracting Officer will obtain services of Naval Facilities Engineering Systems Command (NAVFAC) QEI Certified Elevator Inspector. NAVFAC QEI will utilize the applicable NAVFAC Elevator Acceptance Inspection Form to record the results of inspection and testing and to identify safety code

and contract deficiencies. Provide specific values for all tests required by ASME A17.1/CSA B44, ASME A17.2, and contract documents. Upon completion of inspection and testing, the NAVFAC QEI will sign a copy of the completed forms and provide the signed copy to the Contracting Officer or representative. Within 2 weeks of the inspection, the QEI will also submit an Elevator Certification Certificate, Assessment, or Validation and Compliance Report including all test results and deficiencies. Upon successful completion of inspection and testing, NAVFAC Certified Elevator Inspector will complete, sign and post form NAVFACENGCOM 9-11014/23(Rev.9-2009), Elevator Inspection Certificate.

][Contracting Officer will obtain the services of a third-party QEI Certified Elevator Inspector. The QEI must utilize an Elevator Acceptance Inspection Form to record the results of inspection and all testing and to identify safety code and contract deficiencies. Provide specific values for all tests required by ASME A17.1/CSA B44, ASME A17.2, and contract documents. Upon completion of inspection and testing, the QEI must sign a copy of the completed forms and provide to the Contracting Officer. Within 2 weeks of the inspection, the QEI must also prepare a formal inspection report, including all test results and deficiencies. Upon successful completion of inspection and testing, the QEI will complete, sign, and submit an Elevator Certification Certificate, Assessment, or Validation and Compliance Report with ASME A17.1/CSA B44.

3.3.1 Acceptance Inspection Support

Prime and Elevator Contractors must provide inspection support and perform all required tests, in order to demonstrate proper operation of each elevator system and to prove that each system complies with contract requirements and all applicable building and safety codes. Inspection procedures in ASME A17.2 form a part of this inspection and acceptance testing. Conduct all inspection and testing in the presence of the Qualified Elevator Inspector (QEI).

If the elevator does not comply with all contract and safety code requirements on the initial Acceptance Inspection and Test, the Contractor is responsible for all costs involved with re-inspection and re-testing required as a result of contractor delays and discrepancies discovered during inspection and testing.

3.3.2 Testing Materials and Instruments

Provide all testing materials and instruments necessary for Acceptance Inspection, Testing and Commissioning. At a minimum, include calibrated test weights, tachometer, accelerometer, hydraulic pressure gauge, 600-volt megohm meter, volt meter and ammeter, infrared temperature gauge, door pressure gage, dynamometer, and 6 meter 20 foot tape measure.

3.3.3 Field Tests

3.3.3.1 Endurance Tests

Test each elevator for a period of one hour continuous, automatic operation, with specified rated load in the elevator cab. During the one hour test, stop car at each floor, in both directions of travel, and allow automatic door open and close operation. The requirements for Automatic Operation, Rated Speed, Leveling, Temperature Rise and Motor Amperes must be met throughout the duration of the Endurance Test. Restart the one hour test period from the beginning, following any shutdown or failure.

3.3.3.2 Speed Tests

Determine actual speed of each elevator, in both directions of travel, with rated load and with no load in elevator car. Make Speed tests at the beginning and at the end of the Endurance test. Determine speed by tachometer reading or accelerometer, excluding accelerating and slow-down zones. Under all conditions, minimum acceptable elevator speed is the Rated speed specified. Maximum acceptable elevator speed is 110 percent of Rated speed.

3.3.3.3 Leveling Tests

Test elevator car leveling operation and provide a leveling accuracy equal to or less than 3 mm 1/8 inch at each floor with no load in car, balanced load in car, and with rated load in car, in both directions of travel. Determine leveling accuracy at the beginning and at the end of the endurance tests.

3.3.3.4 Temperature Rise Tests

Determine temperature rise of elevator drive machine motor during one-hour full-load test run. Under these conditions, maximum temperature rise must not exceed acceptable temperature rise indicated on manufacturer's data plate. Start test only when equipment is within 5 degrees C of ambient temperature.

3.3.3.5 Balanced Load Test

Place balanced load in the elevator cab, according to the manufacturer's designed counterbalance. Perform electrical and mechanical balanced load tests of car and counterweight.

3.3.3.6 Motor Ampere Tests

At beginning and end of Endurance test, measure and record motor amperage in both directions of travel and in both no-load and rated load conditions.

3.3.3.7 Elevator Performance and Ride Quality Testing

Evaluate elevator performance to ensure compliance with specification requirements related to the NEII-1 Performance Standards Matrix for New Elevator Installations.

3.3.3.8 Sump Pump Testing

Perform OEM required testing of the installed sump pump and control/alarm system in the presence of the QEI. Minimum testing to include motor amperage, actual flow rate, discharge verification, and alarm operation.

-- End of Section --